Numbering Pages in xslTNG printed media

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Dedication

An dedication could be placed here

Preface

This book examines the handling of page numbers in CSS printed media. It is both documentation and its own illustrative material. It is created in DocBook format and converted to HTML with xslTNG 2.2. It is then rendered to PDF using various rendering engines. The aim is to illustrate the problems that can occur with page numbers in the created PDF and how these can be solved.

Some of the problematic effects are only recognizable when sections in the generated PDF contain a page break. For this reason, some sections of this book are more extensive and detailed than is absolutely necessary. It is expressly intended to fill pages, because we need page breaks for illustration. Also, it's the first time for me to have a `dedication` and a `colophon` in a book. It's only because I wanted to see them rendered in PDF.

This book is translated with DeepL.com from german.

Chapter 1. Motivation

DocBook [DB 5.2] is a general purpose XML schema for structured documents. It's an OASIS Standard. Yo have to transform them into output formats like HTML, PDF or ODF before the audience can read them. For this, you have to use stylesheets.

DocBook: xslTNG are the next generation of DocBook stylesheets following on from the *DocBook XSLT 2.0 Stylesheets* and the *DocBook XSLT 1.0 Stylesheets* that preceded them. They transform DocBook XML documents into clean, semantically rich HTML5. A CSS stylesheet suitable for online presentation is included.

Producing high quality print output with a paged media capable CSS processor is also supported.This is explained in Section 2.1.4 of *xslTNG Ref*, which says: “At present, the *DocBook xslTNG Stylesheets* are focused on CSS for print output. There’s a customization layer that produces “paged-media-ready” HTML that can be processed with a CSS formatter such as Antenna House or Prince”.

Both are commercial products. This creates a high barrier to entry for all those who want to create not only HTML but also PDF with the stylesheets. I am convinced that it is therefore necessary to support CSS paged media processors that are avaliable for free[1].

You can find an excellent tutorial that shows how to generate PDF documents from XML/HTML using the "CSS Paged Media" approach at print-css.rocks. It's not only a tutorial, they do also test various CSS Paged Media converters . WeasyPrint (weasyprint.org/) is a free and open source software for rendering HTML into PDF, so I gave it a try. The result was promising, but there was a problem: references to pages (for example in the table of contents, but also in cross-references) do not show the correct page number, but always a 0 (zero). I found Issue 433 which complains about the same problem. It already identified `counter-reset: page` as problem (“Apparently weasyprint doesn't like that”), but was closed with an only half-hearted solution: `ounter-reset: none` leads to correct page numbers, but the first page for content in the book (after the *Titlepage*, the *Table(s) of Content* and other stuff like that) does not start with page number “1”.

This led to an error message at weasyprint (Issue 2114). The developer provides a comprehensible explanation and a solution that delivers the desired result with weasyprint. However, the solution does not work properly with Antenna House. This doesn't look like an bug to me. On the contrary, Weasyprint's approach seems to me to be problematic in conjunction with CSS “named pages”, so that I can understand Antenna House's result better than Weasyprint's.

However, I am anything but an expert in CSS paged media. I am not in a position to decide which implementation is right and which is wrong. Therefore, I have decided to document both approaches with the corresponding results as well as possible, so that an optimal solution for the xslTNG stylesheets can be developed. In addition to Antenna House and Weasyprint, the product PDF Chemistry from Syncro Soft will also be examined. It is part of the Oxygen product family and can be used free of charge by users of the Oxygen IDE to convert HTML+CSS paged media into PDF. PDF Chemistry is a CSS Paged Media processor based on the open-source Apache FOP XSL-FO engine. In other words, in transforms HTML+CSS into XSL Formatting Objects (XSL-FO) which can be rendered into PDF with the open source FOP Software from Apache, which also ships with Oxygen.

[1] I do believe that most technical documentation should be avalilable in both formats: Online and PDF.

Chapter 2. Test approach and results

1. This Report is the Test-Document

This report here serves as the base HTML Document for the test of CSS Sytlesheets and rendering engines. It is written in DocBook 5.1. The xslTNG Stylesheets Version 2.2 with `print.xsl` are used to produce the HTML file `report.html`.

Parameter `page-style` is set to `book`.

2. Rendering Engines

IThe test was performed with these software products:

- **WeasyPrint version 54.1** on Ubuntu Linux. It is used as

```
weasyprint -s css-file report.html report.pdf
```

- **PDF Chemistry, part of XML Editor 26.0** on Ubuntu Linux. To use it within Oxygen, I had to define “transformation scenarios” of type “XML to PDF transformation with CSS” which refers the `css-file`.

- **Antenna House Formatter 7.4R1** on Windows 10. It is used as

```
ahfcmd.exe -css css-file -d report.html -o report.pdf
```

3. xslTNG CSS

xslTNG brings it's own CSS stylesheets for HTML as well as PDF. When the parameter `use-docbook-css` is set to `true` (the default), the HTML file will refer to CSS like this:

```
<link href="./css/docbook.css" rel="stylesheet" media="screen" />
<link href="./css/docbook-paged.css" rel="stylesheet" media="print" />
```

3.1. Weasyprint

Weasyprint produces a PDF file which looks fine at the first glance, but all page numbers in the ToC are 0 (Zero):

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Figure 2.1. ToC with Weasyprint and the original xslTNG CSS

3.2. PDF Chemistry crashes with NPE

Oxygen's PDF Chemistry complains about `invalid 'target-counter' function signature. Must be target-counter( <url-fragment> , <custom-ident> [ , <counter-style>` with a Warning. It crashes with this message: `java.lang.NullPointerException: Cannot invoke "ro.sync.ecss.css.functions.kb.l(String)" because the return value of "ro.sync.ecss.css.functions.ab.c()" is null`.

Chapter 3. Conclusion

Bibliography

[db 5.2] *DocBook 5.2: The Definitive Guide* by Norman Walsh. Online available at tdg.docbook.org/tdg/5.2/

[xslTNG] *DocBook xslTNG* by Norman Walsh. Online available at xsltng.docbook.org/

[xslTNG Ref] *DocBook xslTNG Reference* by Norman Walsh. Online available at xsltng.docbook.org/guide/

Why you should read this Book (Colophon)

It is written in DocBook 5.1. Translated to HTML with xslTNG 2.2.

Rendered in PDF with different engines:

1. Antenna House
2. WeasyPrint
3. PDF Chemistry (part of the Oxygen XML Suite)